

MANUSHREE SHAH

AI Product Engineer | GenAI & LLM Systems | Full-Stack AI Development

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PROFESSIONAL SUMMARY

AI Product Engineer with 1.8+ years hands-on experience building and scaling AI-first SaaS products, including 0→1 startup environments. Specialization in translating business and user needs into production-ready AI solutions through collaboration with leadership and cross-functional teams at the intersection of product and engineering to deliver scalable, high-impact systems that continuously evolve and drive sustained value. Led end-to-end ownership of AI features — from ideation to deployment — including speech-to-text and LLM-driven workflows, RAG-based chatbots, and autonomous AI agents. Skilled in Python, LangChain, LangGraph, PostgreSQL, Flask, GCP, and LLM Ops pipelines, managing the full development lifecycle for both internal and client-facing products.

TECHNICAL SKILLS

Programming: Python

LLM & Generative AI: LangChain, LangGraph, RAG Pipelines (Pinecone, FAISS), Prompt Engineering (Few-shot, Chain-of-Thought, Self-Consistency), LLM Fine-Tuning (QLoRA, LLaMA), LLM Ops (Versioning, Evaluation, Monitoring)

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, TabNet, Ensemble Models, HuggingFace Transformers, SMOTE

Multimodal AI: Speech-to-Text (STT), Text-to-Speech (TTS), Embeddings, Diffusion Models (Stable Diffusion)

Backend & Data Systems: Flask (REST APIs), PostgreSQL, Data Mining, BI & Visualization (Excel)

Cloud & Infrastructure: Google Cloud Platform (GCP), VM Provisioning, NGINX, GitHub CI/CD, Azure (AI Fundamentals)

Computer Vision: YOLO, Roboflow

PROFESSIONAL EXPERIENCE

Investigen.ai — AI Product Engineer (Jul 2025 – Present)

KEY PROJECTS

Architected and delivered end-to-end production GenAI systems across multiple client-facing financial applications, designing scalable architectures leveraging LLMs, vector embeddings, and RAG pipelines from POC to deployment:

- Built and deployed a **full-stack voice-enabled AI sales coaching platform** (STT/TTS + LLMs) to analyze live sales pitches and delivered actionable feedback — improving pitch consistency by ~30% for 100+ sales professionals at a large AMC, while enabling MIS based performance monitoring.
- Engineered **conversational and agentic AI workflows** using LangChain, LangGraph, and n8n, orchestrating multi-step LLM pipelines to build bot-based financial calculators capable of handling complex planning scenarios, also deploying scalable backend infrastructure (Flask + NGINX) optimized for handling concurrent multi-user queries and higher request loads.
- Developed a **multimodal GenAI platform** processing text, audio, video, PDFs, and marketing creatives, enabling organizations to query and summarize across 100+ monthly concall meetings while automatically evaluating mutual fund advertisements against 40+ SEBI compliance rules to identify regulatory gaps, reducing manual review effort by 70% and accelerating compliance workflows.

- Applied advanced **prompt engineering techniques** (few-shot, chain-of-thought, self-consistency) and **fine-tuned SLM like Gemma 3 (4B) using QLoRA** for domain-specific financial language understanding, contributing to LLMOps workflows including model versioning, evaluation, and monitoring.
- Owned backend APIs using Flask; deployed full-stack on GCP via VM provisioning, NGINX configuration, and GitHub CI/CD pipelines.

Investigen.ai — AI Developer Intern (Jan 2025 – Jul 2025)

- Built and shipped an AI-powered investment proposal generator, automating manual content creation and significantly reducing turnaround time for client-facing financial proposals (Python).
- Developed a RAG-based financial regulations chatbot using Pinecone and FAISS enabling rapid compliance querying and validated machine learning classification models for scheme performance analysis through five-fold cross-validation.
- Engineered chatbot-based calculators in Python (Flask + PostgreSQL) to enable faster, more consistent investment planning workflows (Gemini, OpenAI and Llama LLMs).

Bandhan AMC — Product Intern (Jun 2024 – Oct 2024)

- Automated daily operational tasks with Python to improve workflow efficiency; performed BI tasks including data mining, wrangling, and visualization to surface actionable sales insights.

RESEARCH & PUBLICATIONS

Evaluating Predictive Maintenance with Multiclass Categorical Data Constraints — Springer, IISU ASSET 2025
Developed deep learning and ensemble predictive maintenance models (TabNet, SMOTE) for rail systems, benchmarking failure-prediction accuracy on sparse categorical-multiclass datasets. (PyTorch)

AI-Enabled Automated Inspection of Reinforced Concrete Cement (RCC) — IEEE Xplore, ICSPCRE-2024, NIT Rourkela

Automated rebar alignment inspection using object detection and computer vision (YOLO, Roboflow, Weights & Biases).

Automating Data Manipulation Tasks in MS Excel: Comparative Analysis of Gemini and Llama LLMs — IEEE Xplore, CISCT-2024

Benchmarked Gemini and Llama models on few-shot and zero-shot code generation tasks for Excel manipulation, evaluating across multiple NLP metrics.

EDUCATION

PG Certification in AI Engineering on Cloud & AIOps

CEC, Indian Institute of Technology, Roorkee (Sep 2025 – Jun 2026)

B.Sc. (Hons) in Artificial Intelligence

SVKM's NMIMS University, Navi Mumbai (Aug 2021 – Jun 2025)

Gold Medalist & Best Student Awardee (CGPA: 3.95/4.0)

ACHIEVEMENTS & CERTIFICATIONS

- **Microsoft Certified: Azure AI Fundamentals** — Issued May 2024
- **Student Council President** — NMIMS NM (Apr 2023 – May 2024): Led a 50+ member team, organizing 15+ AI/DS workshops and hackathons to build a thriving campus AI community.